

FinRelief AI

AI-Powered Debt Relief & Financial Recovery Platform

Project Documentation

Prepared for: Internal Project Reference

Version: 1.0

1. Introduction

FinRelief AI is an AI-powered debt relief and financial recovery platform designed to help individuals who are struggling with multiple loans. Many borrowers do not know how to negotiate with banks, how much to realistically offer for a settlement, or what their legal rights are when dealing with recovery agents. FinRelief AI solves this problem by combining a borrower's personal financial data with generative AI to produce a specific, actionable debt-recovery strategy.

The platform allows a user to securely log in, add and manage all of their loans in one place, receive an AI-generated settlement plan for each loan, generate a ready-to-send negotiation letter, and learn about their legal rights as a borrower in India. Every recommendation is personalized — the AI reads the user's own income, expenses, and loan details before generating any output.

2. Project Description

FinRelief AI is a full-stack web application that gives borrowers a centralized, secure dashboard to understand and act on their debt. Instead of manually tracking loans across multiple lenders and guessing at a settlement offer, the platform automatically calculates financial health indicators and uses Google Gemini AI to turn raw loan data into a clear, actionable recovery plan.

The system focuses on three outcomes for the user: clarity on their overall debt position, a realistic settlement strategy for every loan, and confidence in dealing with lenders through AI-generated negotiation letters and legal-rights information.

Key Features

- Secure user registration and login protected by JWT-based authentication.
- Centralized dashboard showing monthly surplus, total outstanding debt, total EMI, debt-to-income ratio, and an overall financial stress level.
- Add, view, and manage multiple active loans (personal loan, home loan, car loan, etc.) with details such as sanctioned amount, outstanding balance, interest rate, EMI, overdue months, and due date.
- Automatic loan prioritization (High / Medium / Low) based on interest rate and overdue period.
- AI-powered Settlement Predictor that recommends a one-time settlement amount (typically 40%–75% of the outstanding balance) for each loan, based on its risk category.
- AI-generated Negotiation Email that drafts a formal settlement letter to the lender, including the hardship situation, offer amount, penalty-interest waiver request, and a No Objection Certificate (NOC) request.
- "Know Your Rights" section explaining RBI guidelines on fair recovery practices and how to escalate complaints to the Banking Ombudsman.
- Private History log that automatically stores every AI-generated strategy and letter for the logged-in user only.

3. Scenario-Based Case Study

Scenario: Fathima, a salaried employee, has taken multiple loans over the past few years — including a personal loan from a digital lender — and is finding it difficult to keep up with the combined EMIs. She decides to use FinRelief AI to get a clear picture of her debt and a practical way forward.

Step 1 — Registration and Login

- Fathima opens the FinRelief AI website and signs in with her registered email and password.
- The backend verifies her credentials and issues a JWT token, which is used automatically on every subsequent page so she stays securely logged in.

Step 2 — Setting Up Her Financial Profile

- From the Dashboard, Fathima clicks "Edit Profile" and enters her monthly income, monthly expenses, and any lump sum amount she has available for a settlement.
- The dashboard instantly recalculates her monthly surplus and debt-to-income ratio using this data.

Step 3 — Adding Her Loans

- She clicks "Add Loan" and enters the loan type, lender name, sanctioned amount, outstanding balance, interest rate, EMI, overdue months, and due date for each active loan.
- Each loan immediately appears in her Active Loans list along with a calculated priority level, so she can see at a glance which loan needs urgent attention.

Step 4 — Getting an AI Settlement Plan

- Fathima opens the Settlement Predictor. The system evaluates each loan's risk and recommends a realistic one-time settlement amount.
- Google Gemini AI reviews all of her loans together and explains which loan to prioritize and how much she could save overall by settling instead of continuing full repayment.

Step 5 — Generating a Negotiation Letter

- She selects her most urgent loan in the Negotiation Email section and generates a formal settlement letter addressed to the lender, ready to send without needing a lawyer.

Step 6 — Knowing Her Rights

- Before contacting her lender, Fathima reviews the "Know Your Rights" page to understand what recovery agents are and are not allowed to do, giving her confidence during the negotiation.

Step 7 — Reviewing History

- All the strategies and letters generated for Fathima are saved in her private History page, so she can revisit them anytime without regenerating them.

4. Technical Architecture

FinRelief AI follows a client-server architecture in which a React.js single-page application acts as the client and a Python FastAPI service acts as the server. The frontend communicates with the backend over RESTful APIs using Axios, and every protected request carries a JWT token that identifies the logged-in user.

Layer	Technology	Purpose
Frontend	React.js (Vite) + Axios	User interface, dashboard, forms, and API calls
Backend	FastAPI (Python)	REST API, business logic, authentication, routing
Database	SQLite + SQLAlchemy	Stores users, loans, settlement records, and AI history
AI Engine	Google Gemini 2.5 Flash	Generates settlement strategies and negotiation letters
Security	JWT (HS256) + password hashing	Authenticates users and protects private data

The backend is organized into modular routers (authentication, loans, settlement prediction, negotiation email, financial health, AI history, and rights information). Each protected endpoint filters data by the current user's ID, ensuring one borrower can never view another borrower's loans or history. When the Gemini API is temporarily unavailable or a rate limit is hit, the AI engine automatically falls back to a rule-based template so the user always receives an output.

5. Technology Stack & Prerequisites

Backend

- **FastAPI** — modern, high-performance Python web framework used to build the REST API.
- **SQLite with SQLAlchemy** — lightweight relational database and ORM for storing users, loans, settlement records, and AI history.
- **JWT (JSON Web Tokens)** — used for secure, stateless authentication; tokens expire after 24 hours.
- **Password hashing** — user passwords are never stored in plain text.
- **Google Gemini 2.5 Flash API** — generates personalized settlement strategies and letters.

Frontend

- **React.js (Vite)** — component-based UI library with a fast development build tool.
- **Axios** — handles all HTTP communication between the frontend and backend.

Development Tools

- Visual Studio Code as the primary code editor.
- Git for version control.

- Uvicorn as the ASGI server used to run the FastAPI backend locally.

6. Backend Module Structure

The backend is split into a main entry point and eight functional modules, keeping authentication, financial logic, and AI logic cleanly separated.

File / Module	Responsibility
app/main.py	Application entry point; registers all routers and configures CORS
app/core/logic.py	Core financial calculations — EMI ratio, stress level, monthly surplus, settlement percentages
app/core/ai_engine.py	Builds the AI prompt from loan data, calls Google Gemini, and applies a rule-based fallback if the
app/routers/auth.py	User registration, login, password hashing, and JWT token creation
app/routers/loans.py	Add, update, and retrieve loan records for the logged-in user
app/routers/settlement.py	Settlement Predictor endpoint — risk scoring and recommended offer
app/routers/negotiation.py	Generates the AI-powered negotiation letter for a selected loan
app/routers/ai_history.py	Stores and retrieves each user's AI-generated results, filtered by user ID for privacy
app/routers/rights.py	Serves the "Know Your Rights" borrower-education content

FastAPI also auto-generates interactive API documentation at the **/docs** endpoint, listing all routes across the eight modules and clearly marking which endpoints require a valid JWT token.

7. Application Flow

User Role: Borrower — the platform is designed around a single user role, the borrower, who manages their own debt end-to-end.

- 1. Registration and Login:** The borrower creates an account and logs in; the backend issues a JWT token that authenticates every following request.
- 2. Financial Profile Setup:** The borrower enters monthly income, monthly expenses, and any lump sum available, forming the baseline used for all calculations.
- 3. Loan Management:** The borrower adds each active loan with its type, lender, amount, interest rate, EMI, and overdue status, and can view or remove loans at any time.
- 4. Dashboard Review:** The system calculates monthly surplus, total outstanding debt, total EMI, debt-to-income ratio, and a stress level (Low / Medium / High), and ranks loans by priority.
- 5. Settlement Prediction:** The borrower runs the Settlement Predictor, which evaluates every loan's risk and returns a recommended settlement percentage along with an AI-written analysis of which loan to prioritize.
- 6. Negotiation Letter Generation:** The borrower selects a loan and generates a formal, ready-to-send settlement letter to the lender.

7. Rights Awareness: The borrower reviews RBI recovery guidelines and Banking Ombudsman escalation steps before engaging with the lender.

8. History Tracking: Every AI-generated strategy and letter is automatically saved to the borrower's private History page for future reference.

8. Application Screenshots

Login Page

The Login page presents a dark, professional theme with the platform's value proposition on the left and a sign-in form on the right, allowing a borrower to sign in or register.

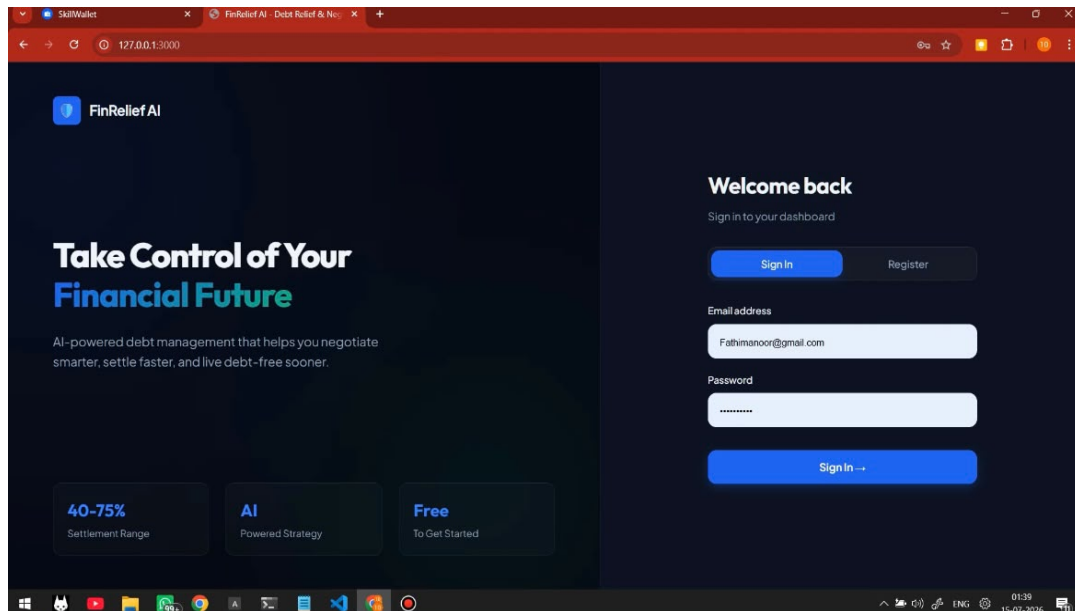


Fig 1: Login Page

Dashboard Overview

The Dashboard shows monthly surplus, total outstanding balance, total EMI, debt-to-income ratio, stress level, the borrower's financial profile, and a ranked list of active loans.

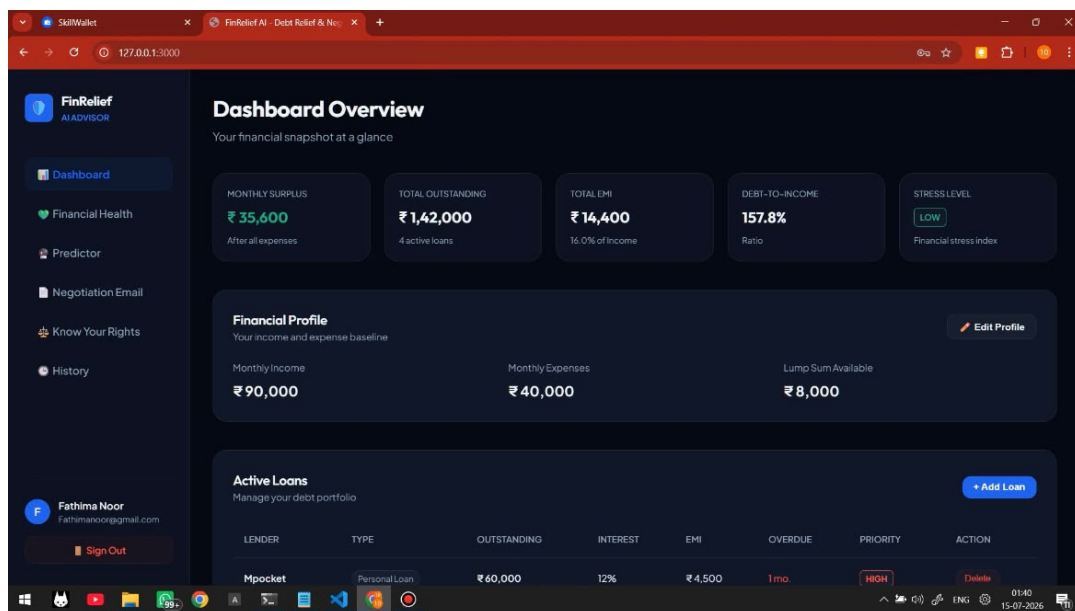
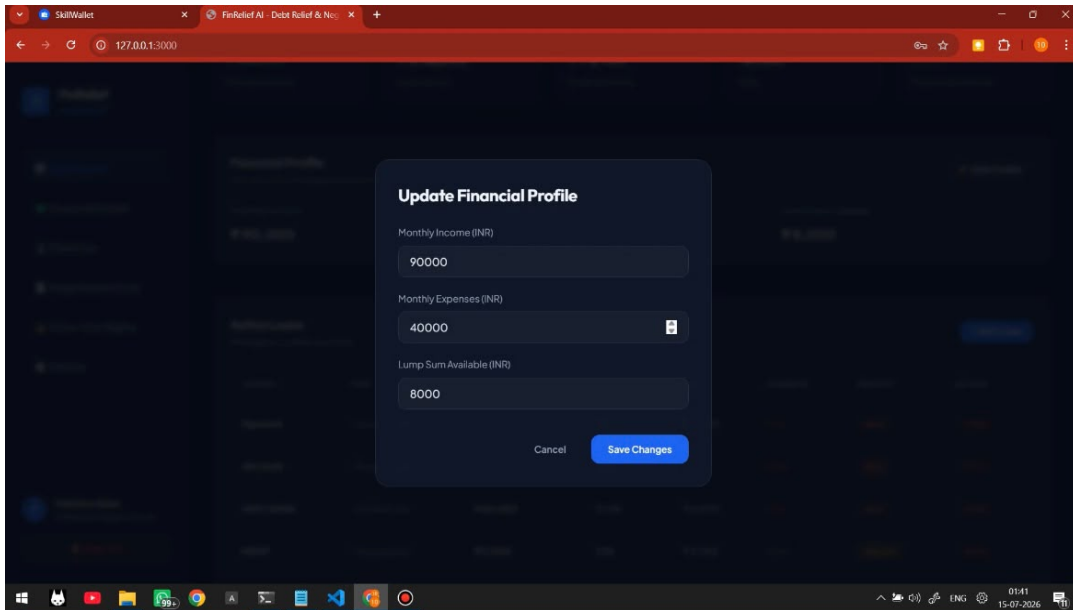


Fig 2: Dashboard Overview

Update Financial Profile

A borrower can edit their monthly income, monthly expenses, and available lump sum at any time; the dashboard recalculates instantly based on the updated values.

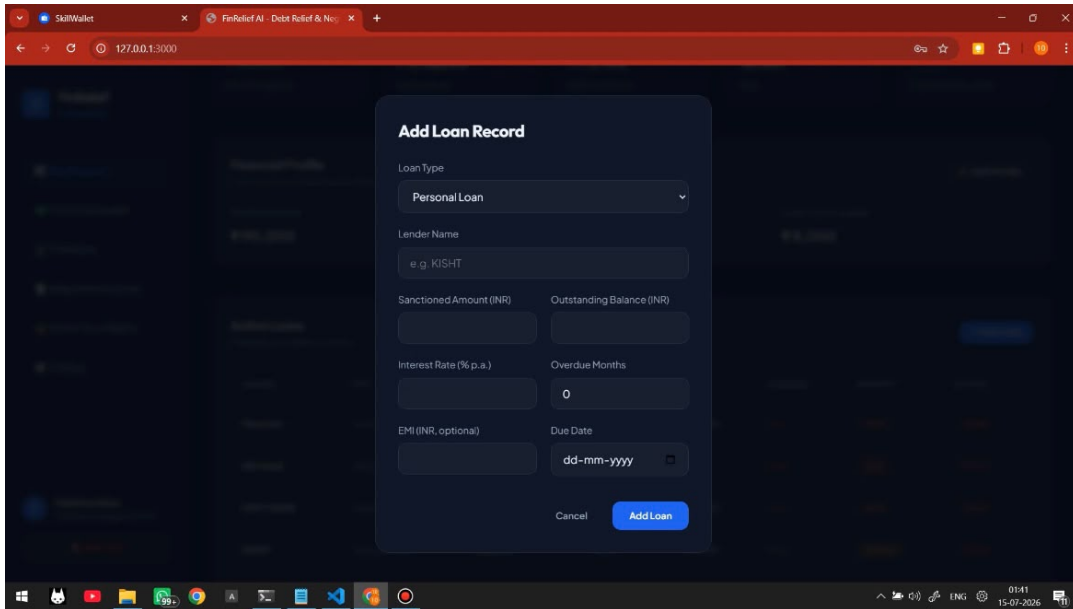


The screenshot shows a web browser window with a dark-themed dashboard. A modal titled "Update Financial Profile" is centered on the screen. It contains three input fields: "Monthly Income (INR)" with the value "90000", "Monthly Expenses (INR)" with the value "40000", and "Lump Sum Available (INR)" with the value "8000". At the bottom of the modal are two buttons: "Cancel" and "Save Changes". The background dashboard is dimmed, showing various financial metrics and charts.

Fig 3: Update Financial Profile

Add Loan Record

New loans are added through a simple form capturing loan type, lender name, sanctioned amount, outstanding balance, interest rate, EMI, overdue months, and due date.



The screenshot shows the same web browser window with the "Add Loan Record" modal open. The form includes a "Loan Type" dropdown menu set to "Personal Loan", a "Lender Name" text field with the placeholder "e.g. KISHT", and two columns of input fields. The left column contains "Sanctioned Amount (INR)", "Interest Rate (% p.a.)", and "EMI (INR, optional)". The right column contains "Outstanding Balance (INR)", "Overdue Months" (set to "0"), and "Due Date" (set to "dd-mm-yyyy"). At the bottom are "Cancel" and "Add Loan" buttons. The background dashboard remains dimmed.

Fig 4: Add Loan Record

9. Summary

FinRelief AI brings together financial calculation logic and generative AI to give borrowers a single, secure place to understand their debt and act on it. With JWT-secured accounts, an eight-module FastAPI backend, a React.js frontend, and Google Gemini-powered strategy and letter generation, the platform turns a stressful, confusing process into a clear, personalized, step-by-step plan — helping borrowers negotiate smarter, settle faster, and move toward a debt-free future.